

Varanasi Sai Shashank

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AI/ML Engineer

Summary

AI/ML engineer and full-stack developer experienced in **building production AI features and scalable backends**. Hands-on with **RAG, LLM integration, transformers**, vector search, and **NLP** for real products. Own end-to-end delivery in startup settings: **requirements** → **system design** → **APIs/data** → **ML integration** → **deploy/monitor**. Strong in Python/Java/JS with a security- and reliability-first mindset.

Technical Skills

Languages	Python, Java, JavaScript
Frameworks/Libs	FastAPI, Flask, Express.js, LangChain, TensorFlow, PyTorch, Scikit-learn, OpenCV, Streamlit
AI/ML	NLP, Transformers, RAG, LangChain, Embeddings, Vector Search, CNN, Random Forest, SVM
Data/Infra	PostgreSQL, Pinecone (Vector DB), AWS S3, Redis, REST APIs, Git, Pandas, NumPy, Dlib
Practices	System Design, API Design, Testing, Code Reviews, Agile, Documentation
DevOps/Cloud	Docker, AWS ECS Fargate, AWS EC2, AWS IAM, AWS Lambda, AWS CloudWatch

Experience

AI/ML Engineer

Jun 2025 — Present

Brillius Technologies, Hyderabad

Product: EdTech AI learning platform for IT professionals — personalized learning, retrieval-augmented Q&A, and assessments.

- **Backend ownership:** Designed and implemented RESTful services for auth, user/course management, content ingestion, analytics, and notifications using **Python (FastAPI/Flask)** and **Node.js (Express.js)**.
- **Data layer:** Modeled relational schemas in **PostgreSQL**, implemented migrations, indexes, constraints, pagination, and safe transactional updates.
- **AI platform integration:** Built **RAG pipelines** using **LangChain** (chunking, transformer embeddings, vector search on **Pinecone**), LLM orchestration with **OpenAI API** and **Ollama**; added prompt templates and guardrails.
- **NLP features:** Implemented document Q&A, summarization, keyword extraction, level classification (Beginner/Intermediate/Expert), and quiz generation workflows.
- **Storage/Infra:** Managed assets in **AWS S3** (signed URLs, lifecycle policies); deployed AI microservices on **AWS ECS Fargate**; reduced latency via **Redis** caching strategies.
- **Collaboration & Leadership:** Managed Git workflows and technical documentation; acted as interim team lead driving timely delivery.

Projects

Driver Monitoring System — Real-time Fatigue & Drowsiness Detection

Jan 2025 — May 2025

Python, OpenCV, Dlib, CNN, DeepFace, EAR, MAR, Head-pose, Tkinter

- Built a live camera pipeline for face/eye landmarking, **EAR & MAR** computation, head-pose estimation, and distraction detection; triggered multi-modal alerts (audio, visual) for micro-sleep and prolonged yawns.
- Integrated **CNN + DeepFace** for expression analysis and added a driving time tracker that issues fatigue warnings after 10 hours, achieving **96% detection accuracy**.

[GitHub](#) | [Publication — SafeDriveVision, IJARESM Vol 13 Issue 4, Apr 2025](#)

Voice Interaction System — TTS/STT + Sentiment Analysis

Jul 2024 — Dec 2024

Python, Google STT API, pyttsx3, VADER, NLTK, Streamlit

- Built an integrated speech pipeline: wake phrase → STT transcription → VADER sentiment analysis (positive/negative/neutral) → TTS response; provides real-time emotion-aware feedback.
- Deployed via **Streamlit** with multilingual support; combined Google Speech Recognition API for transcription with

pyttsx3 for natural-sounding voice output.

[GitHub](#) | [Publication — Converting Text to Speech and STT Using NLP, IJRPR Vol 5 No 11, Nov 2024](#)

Fake Profile Detection — Instagram

Feb 2024 — Jun 2024

Python, Scikit-learn, Random Forest, SVM, APIFY API, Streamlit

- Engineered features from metadata and behavioral signals (post count, followers, following, verified, private); trained SVM, Logistic Regression, and Random Forest — **Random Forest achieved 98% accuracy**.
- Integrated **APIFY Instagram scraper API** for real-time profile prediction; deployed on Streamlit with a scoring API that returns top feature explanations.

[GitHub](#) | [Publication — Fake Profile Detection using ML and Deep Learning, IJRPR Vol 5 No 6, Jun 2024](#)

Education

B.Tech in Artificial Intelligence & Machine Learning

2021 — 2025

Malla Reddy University, Hyderabad

CGPA: **9.07 / 10**

Class XII — AP Board (English Medium)

2021

Narayana Junior College, Ongole

94.3%

Class X — AP Board (English Medium)

2019

Sri Chaitanya High School, Tangutur

GPA: **9.8 / 10**

Certifications

HackerRank Python Developer, Java Basics

Feb 2025

Coursera AI for Everyone, Python for Data Visualization, Intro to HTML5 & CSS3, PHP DB Connectivity

KodeKloud AWS S3

Unstop Hero Campus Challenge Season 8

Cambridge English B2